Activity 4

Angle sum for a polygon

Draw any quadrilateral *ABCD*.

Mark a point *P* in its interior.

Join the segments *PA*, *PB*, *PC* and *PD*.

You have 4 triangles now.

How much is the sum of all the angles of the 4 triangles?

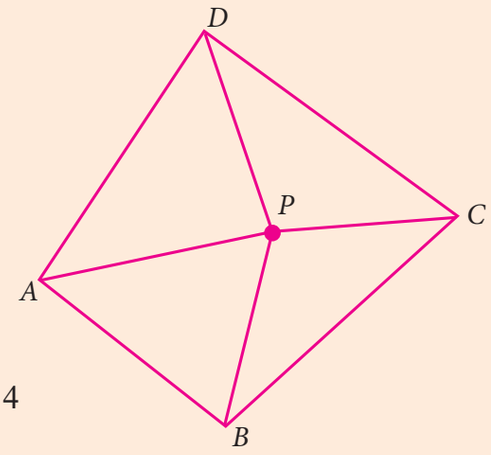


Fig. 4.27

How much is the sum of the angles at P?

Can you now find the 'angle sum' of the quadrilateral *ABCD*?

Can you extend this idea to any polygon?

Thinking Corner

1. If there is a polygon of n sides ($n \geq 3$), then the sum of all interior angles is $(n-2) \times 180°$

2. For the regular polygon (All the sides of a polygon are equal in size)

   - Each interior angle is $\frac{(n-2)}{n} \times 180°$
   - Each exterior angle is $\frac{360°}{n}$
   - The sum of all the exterior angles formed by producing the sides of a convex polygon in the order is 360° .
   - If a polygon has 'n' sides , then the number of diagonals of the polygon is $\frac{n(n-3)}{2}$

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